

Aranya Dutta

Systems & Backend Engineer | C · C++ · Python · Go · Linux · Docker

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COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **Codeforces** Pupil | Max Rating 1234 · Consistent improvement from 800 in 6 months [Profile](#)
- **LeetCode** Contest Rating 1623 · 120+ problems solved [Profile](#)
- **GeeksforGeeks** 450+ problems solved across DSA topics [Profile](#)
- **2nd Runner-up** — ICDCIT Hackathon 2024 (100+ teams competing) [Certificate](#)
- **JLPT N5** — Japanese Language Proficiency Test

PROJECTS

KernelTalk · C · Linux Kernel · IPC · Concurrency [GitHub](#)

- Built a **Linux kernel module** enabling multi-client communication via a custom character device ([/dev/kerneltalk](#))
- Implemented **producer–consumer synchronization** using a **circular buffer** with concurrent read/write.
- Designed **blocking and non-blocking I/O** mechanisms for controlled kernel–user space message flow across multiple processes
- Worked directly with kernel file operations (open, read, write, ioctl) in C.

Network Intrusion Detection System (NIDS) · Python · FastAPI · Docker · Networks [GitHub](#)

- Designed a **microservices pipeline** processing raw **PCAP network traffic** end-to-end: packet capture → TCP/IP flow extraction → ML inference → decision engine
- Orchestrated **5+ independent containerized services** (CICFlowMeter, Preprocessor, Model Service, Decision Engine, API Gateway) via Docker Compose
- Debugged and resolved **cross-container networking, service discovery, and environment config** issues in a live multi-process distributed setup
- Wrote full **technical documentation** for the CICFlowMeter service architecture - [Link](#)

SafeShare Backend · Python · FastAPI · Cryptography · Docker [GitHub](#)

- Built a production-grade **secure file-sharing API** with JWT stateless auth, Bcrypt password hashing, and role-based access control (RBAC) middleware
- Authored full **SRS, System Architecture, and Testing Plans** - documented to industry SDLC standards before writing a line of code

Flamingo Chat Backend · Go · WebSocket · Concurrency [Blog](#)

- Designed and built a **concurrent chat server in Go** leveraging goroutines and channels for simultaneous multi-client WebSocket connections
- Applied **Go concurrency primitives** (goroutines, select, channels) - directly relevant to multithreaded, low-latency system design

EXPERIENCE

Tech Lead · AlgoZenith KIIT Chapter

2025 – Present

- Lead the competitive programming community at KIIT - organize problem-solving sessions, mentor students on DSA and algorithmic thinking
- Designed and maintain the chapter website; coordinate technical workflows across the community

EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Expected May 2027

B.Tech Computer Science & Engineering · Concentration: Backend & Systems Engineering

Relevant: OS, Computer Networks, Distributed Systems, DSA, DBMS, Cloud Computing

SKILLS & TECHNOLOGIES

Languages: C · C++ · Python · Go · SQL · Java

Systems & Concurrency: Linux Kernel Modules · Character Devices · IPC · Multithreading · Blocking/Non-blocking I/O · Circular Buffers

Networking: TCP/IP · PCAP Analysis · Network Flow Extraction · REST APIs · WebSockets · Network Protocols

Infrastructure: Docker · Docker Compose · Microservices · CI/CD (GitHub Actions) · PostgreSQL · Google Cloud

Tools: Git · FastAPI · SQLAlchemy · Pytest · Gradle · Maven · Swagger/OpenAPI

CERTIFICATIONS

Google Cloud: Foundations of Cloud Computing · [Badge](#)